

Title

Realtime Motion Capture as an Automated MoBI Quality Control Mechanism

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Abstract (494/503)**Introduction**

Assessing behavior during multimodal data collection in pediatric mental health research poses significant methodological challenges. Children with neurodevelopmental disorders, including ADHD and ASD, often exhibit increased motor activity, fidgeting, and wandering head movements, complicating eye tracking, physiological signals, and EEG data (Thapar et al., 2015; Church et al., 2010). Without objective behavioral context, it is difficult to determine whether degraded data reflect technical failure, movement-related artifacts, or meaningful within-session variability.

Methods

We developed [zed2i motion classifier](#), an open-source Python pipeline that uses markerless 3D motion capture from a stereo camera to classify participant movement in real time. The system interfaces with the Stereolabs ZED 2i camera to extract 3D skeletal joint positions and local joint orientations at 30Hz without relying on wearable sensors. Body tracking uses the ZED SDK's pretrained models to extract continuous skeletal kinematics, from which per-frame joint features—including displacement, velocity, angular change, and relative motion between adjacent joints—are computed and aggregated over sliding windows to support robust behavioral classification. Our motion classifier outputs three key behavioral classification labels for body movements: a behavioral classification, postural shift, and head direction.

Action classification distinguishes “idle” from “moving” states using the SDK's action inference. If a body is classified as “moving”, our motion classifier will not assign a behavioral classification label. This prevents misleading data interpretation during large-scale movement and isolating fine-grained motor variability from gross repositioning. If a body is classified as “idle”, behavioral classification is calculated using distal upper-limb joints across 15-frame windows. Velocities are computed between consecutive frames, and velocities exceeding 0.01 m/frame are flagged as active movement, a heuristic threshold chosen to capture subtle limb motion while minimizing sensor noise. If at least 5 of the 15 frames exceed this threshold, the joint is considered actively moving. The average velocity across all actively moving joints yields a

continuous fidgeting intensity score, normalized to a 0–1 scale and categorized as slightly fidgety, fidgety, or very fidgety.

Postural shifts are detected using local orientation changes along the torso chain, including the pelvis and spinal segments. Sustained deviations from baseline are identified to distinguish stability from meaningful postural reorientation. Head orientation is estimated using vectors defined by facial and upper-body landmarks, and angular deviations are used to classify cardinal head direction, supporting interpretation of gaze- and attention-dependent signals commonly used in MoBI paradigms.

The pipeline streams continuous per-frame and windowed behavioral labels to Lab Streaming Layer (LSL) for synchronization with multimodal data. Synchronized recordings are also saved, enabling offline reprocessing and behavioral classification refinement.

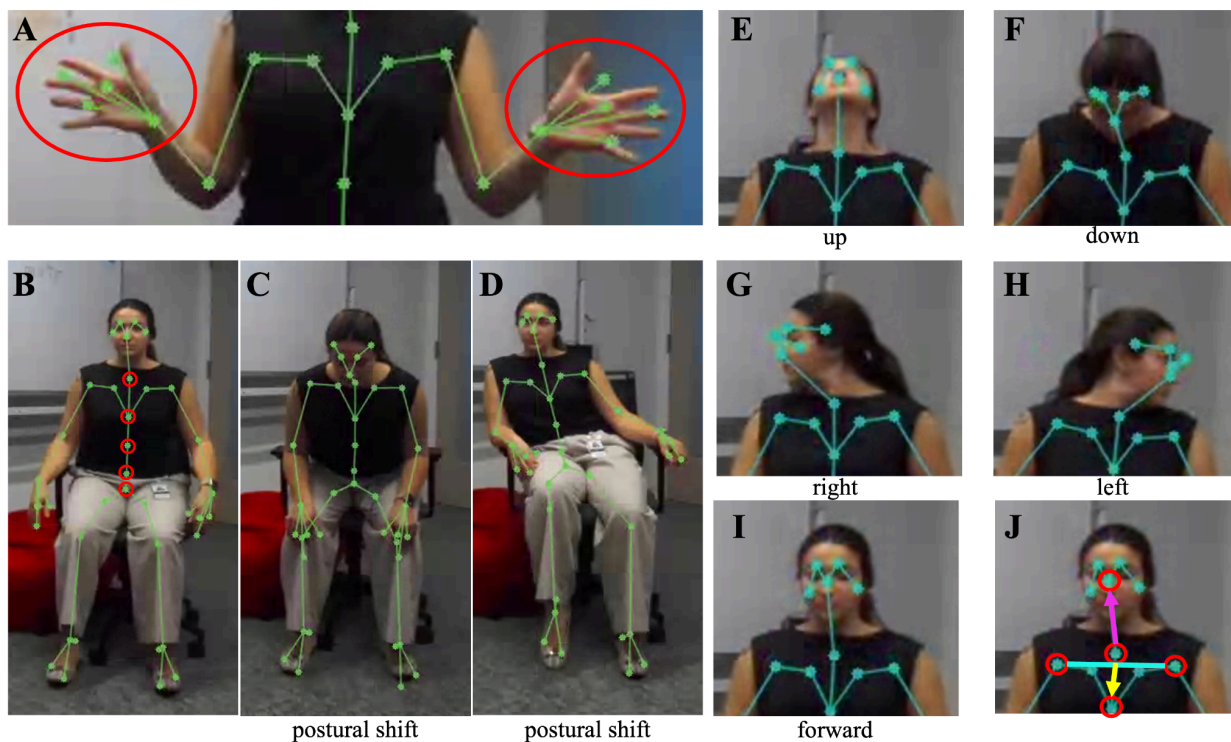
Discussion

zed2i_motion_classifier enables objective, continuous movement-based quality control by providing continuous behavioral signals alongside physiological measures. Researchers can identify movement-affected segments, adjust protocols, or stratify data based on movement artifacts. This allows natural movement during testing while preserving interpretability. Motion features can serve as markers of postural and attentional control, capturing meaningful behavioral phenotypes when combined with neurocognitive measures (Wilson et al., 2024). This tool addresses a critical methodological gap by enabling real-time, non-invasive behavioral monitoring in MoBI research.

Figure 1

Markerless 3D motion tracking and derived features using the *zed2i_motion_classifier*.

Example outputs from the *zed2i_motion_classifier* pipeline illustrating markerless 3D body tracking and derived movement features. (A) Upper-body skeletal tracking with detailed hand joint estimation, demonstrating capture of fine-grained limb and finger movements used for classifying behavior (fidgeting). (B–D) Seated postural variability, including neutral posture (B), forward flexion (C), and lateral leaning with limb repositioning (D), highlighting frame-by-frame changes in joint positions used in classifying postural shifts. (E–I) Head and upper-body orientation changes, including upward tilt (E), downward tilt (F), lateral head rotations (G–H), and forward neutral position (I). (J) Schematic illustrating the vectors derived from facial and torso landmarks (e.g., nose, shoulders, and upper spine) used to compute head orientation. Together, these panels demonstrate the system’s ability to capture continuous, non-invasive measures of movement intensity, postural stability, and head orientation in real time for synchronization with multimodal physiological recordings.



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